

Polar Unit Test (Practice B)

Calculator

You have the entire period.

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test. _____

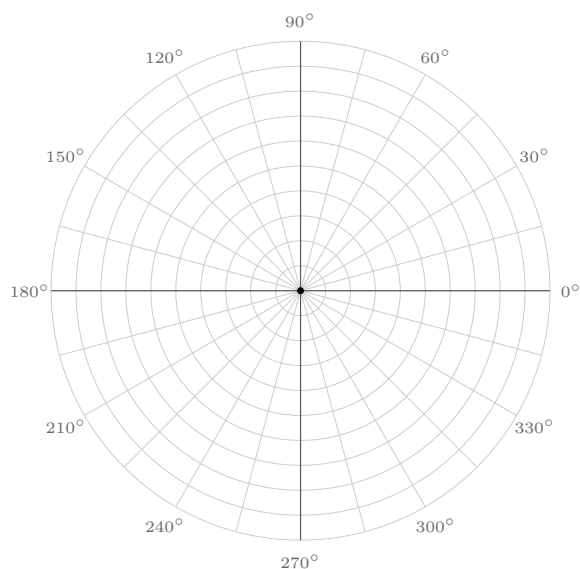
SHOW ALL WORK FOR FULL CREDIT.

Signature _____

Use your calculator to sketch a graph of each polar equation. Identify the type of polar graph it is. Show your set up (what you are putting into the calculator). (3 points each)

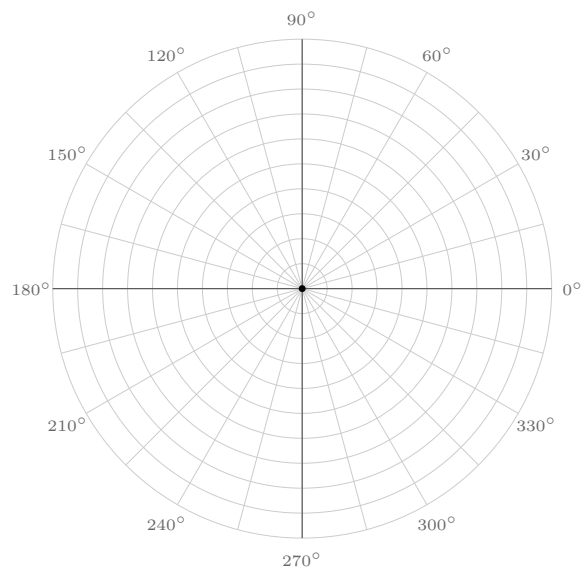
Word bank: circle cardioid limaçon rose curve lemniscate

1. $r = 5 - 5 \cos \theta$



Type: _____

2. $r = 4 \sin 3\theta$



Type: _____

Polar Unit Test (Practice B) *Non-Calculator***SHOW ALL WORK FOR FULL CREDIT.****Simplify. (#3 is 3 points and #4 is 2 points)**

3. $\frac{\sqrt{15}}{\sqrt{-3} \cdot \sqrt{-10}}$	4. $\frac{-10 + \sqrt{-75}}{15}$
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Perform the indicated operations. Write answers in rectangular form. (2 points each for 5 & 6 and 3 points each for 7 & 8)

5. i^{146}	6. $(7 - 2i) - (-3 - 5i) - (12 + i)$
7. $\frac{\sqrt{12} - i\sqrt{12}}{\sqrt{3} + i\sqrt{3}}$	8. $-4i(2 + i)(3 - 2i)$

Solve each equation over the set of complex numbers. (3 points)

9. $3x^2 + 2 = 2x$

Convert to trig form. (3 points)

10. $-3\sqrt{3} + 3i$

Convert to rectangular form. (3 points)

11. $4 \operatorname{cis} \frac{5\pi}{4}$

Using $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$ is the following in the Julia set? Show ALL work. (3 points)

12. $z = -1$

YES / NO

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

13. $\frac{5\sqrt{2}(\cos 60^\circ + i \sin 60^\circ)}{10(\cos 285^\circ + i \sin 285^\circ)}$

14. $[\sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})][4(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2})]$

Using $z, z^2 + z, (z^2 + z)^2 + z, \dots$ is the following number in the Mandelbrot set? Show ALL work. (3 points)

15. $z = 1 + i$

YES / NO

Find each power. Write your answer in rectangular form. (3 pts)

16. $(-1 - i)^6$

Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

17. $x^4 - (-8 + 8i\sqrt{3}) = 0$

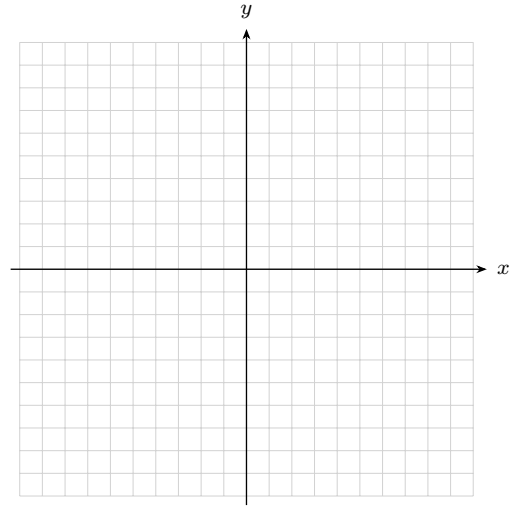
Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points) 18. $(-8, \frac{3\pi}{4})$	Convert the rectangular coordinates to polar coordinates with $r > 0$ and $0^\circ \leq \theta < 360^\circ$. (3 points) 19. $(3, -3\sqrt{3})$
Convert the rectangular equation to a polar equation. Solve for r. (3 points) 20. $5x + 2y = 8$	Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points) 21. $(-3, 0^\circ)$ and $(6, 90^\circ)$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

22. $r = -8 \sin \theta$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

23. $r = \frac{5}{\cos \theta + \sin \theta}$



Find the polar coordinates of the points of intersection for the given interval of θ . (4 points)

24. $r = 3 + 2 \cos \theta$, $r = 3 - 2 \sin \theta$
 $0 \leq \theta < 2\pi$

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

25. $r^2 = 16 \cos 2\theta$

Type: _____

ANSWER KEY – Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Pace check: about 50 minutes for a strong pass. Calculator part about 5, #3–12 about 15, #13–17 about 13, #18–25 about 15. That leaves about 30 minutes of the period to check your work.

Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

1. $r = 5 - 5 \cos \theta$

WORK

Polar mode: $r_1(\theta) = 5 - 5 \cos \theta$

$\theta = 0^\circ: r = 5 - 5 = 0$

$\theta = 90^\circ, 270^\circ: r = 5$

$\theta = 180^\circ: r = 5 - 5(-1) = 10$

$\frac{a}{b} = \frac{5}{5} = 1$

cardioid

WHY

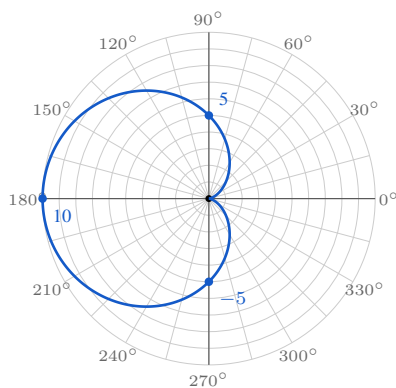
Degree mode needs θ from 0 to 360, or nothing shows.

Touches the pole: the dent of the heart.

Crosses the y -axis at 5 and -5 .

Farthest point, $(-10, 0)$.

Ratio exactly 1.



Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

2. $r = 4 \sin 3\theta$

WORK

Polar mode: $r1(\theta) = 4 \sin(3\theta)$

$n = 3$ (odd): 3 petals

$\theta = 30^\circ$: $r = 4 \sin 90^\circ = 4$

tips at 30° , 150° , 270°

rose curve

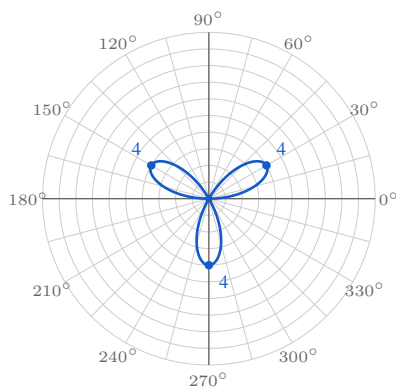
WHY

Type the parentheses around 3θ .

n odd: n petals. n even: $2n$.

Petal length = $|A| = 4$.

At 90° , $r = -4$ plots at 270° .



ANSWER KEY – Non-Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Simplify. (3 points)

3. $\frac{\sqrt{15}}{\sqrt{-3} \cdot \sqrt{-10}}$

WORK

$$\begin{aligned}\frac{\sqrt{15}}{i\sqrt{3} \cdot i\sqrt{10}} &= \frac{\sqrt{15}}{i^2\sqrt{30}} \\ &= \frac{\sqrt{15}}{-\sqrt{30}} = -\sqrt{\frac{15}{30}} = -\frac{1}{\sqrt{2}} \\ &= -\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \boxed{-\frac{\sqrt{2}}{2}}\end{aligned}$$

WHY

Pull out i first, then multiply.

$i^2 = -1$; one radical, reduce inside.

Rationalize.

Trap: Pull out i before multiplying. Writing $\sqrt{-3} \cdot \sqrt{-10} = \sqrt{30}$ flips the sign: $+\frac{\sqrt{2}}{2}$ is wrong.

Simplify. (2 points)

4. $\frac{-10 + \sqrt{-75}}{15}$

WORK

$$\begin{aligned}\frac{-10 + 5i\sqrt{3}}{15} \\ = -\frac{10}{15} + \frac{5\sqrt{3}}{15}i = \boxed{-\frac{2}{3} + \frac{\sqrt{3}}{3}i}\end{aligned}$$

WHY

$\sqrt{-75} = i\sqrt{25 \cdot 3} = 5i\sqrt{3}$.

Split, then reduce each fraction by 5.

Trap: Reduce both parts by the same factor; the i part keeps its own coefficient.

Perform the indicated operation. Write the answer in rectangular form. (2 points)

5. i^{146}

WORK

$$146 \div 4 = 36 \text{ R } 2$$

$$i^{146} = i^2 = \boxed{-1 + 0i}$$

WHY

Only the remainder matters.

Perform the indicated operations. Write the answer in rectangular form. (2 points)

6. $(7 - 2i) - (-3 - 5i) - (12 + i)$

WORK

$$7 - 2i + 3 + 5i - 12 - i$$

$$\text{real: } 7 + 3 - 12 = -2; \text{ imaginary: } -2 + 5 - 1 = 2$$

$$\boxed{-2 + 2i}$$

WHY

Distribute each minus sign to both parts.

Trap: $-(12 + i)$ is $-12 - i$: both signs flip.

Perform the indicated operation. Write the answer in rectangular form. (3 points)

7. $\frac{\sqrt{12} - i\sqrt{12}}{\sqrt{3} + i\sqrt{3}}$

WORK

$$\frac{\sqrt{12} - i\sqrt{12}}{\sqrt{3} + i\sqrt{3}} \cdot \frac{\sqrt{3} - i\sqrt{3}}{\sqrt{3} - i\sqrt{3}}$$

top: $\sqrt{36} - i\sqrt{36} - i\sqrt{36} + i^2\sqrt{36} = 6 - 12i - 6 = -12i$

bottom: $3 + 3 = 6$

$$\frac{-12i}{6} = \boxed{0 - 2i}$$

WHY

Multiply by the conjugate of the bottom.

$$\sqrt{12}\sqrt{3} = \sqrt{36} = 6.$$

$$a^2 + b^2.$$

Perform the indicated operations. Write the answer in rectangular form. (3 points)

8. $-4i(2 + i)(3 - 2i)$

WORK

$$(-8i - 4i^2)(3 - 2i) = (4 - 8i)(3 - 2i)$$

$$= 12 - 8i - 24i + 16i^2$$

$$= 12 - 32i - 16 = \boxed{-4 - 32i}$$

WHY

Distribute the $-4i$ first; $-4i^2 = +4$.

FOIL.

Solve each equation over the set of complex numbers. (3 points)

9. $3x^2 + 2 = 2x$

WORK

$$3x^2 - 2x + 2 = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4(3)(2)}}{2(3)} = \frac{2 \pm \sqrt{-20}}{6}$$

$$= \frac{2 \pm 2i\sqrt{5}}{6}$$

$$\boxed{x = \frac{1}{3} \pm \frac{\sqrt{5}}{3}i}$$

WHY

Zero on one side first.

$$a = 3, b = -2, c = 2; -b = +2.$$

$$\sqrt{-20} = 2i\sqrt{5}.$$

Reduce both parts.

Convert to trig form. (3 points)

10. $-3\sqrt{3} + 3i$

WORK

$$r = \sqrt{27 + 9} = 6$$

$$\theta = \tan^{-1}\left(\frac{3}{-3\sqrt{3}}\right) = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -30^\circ$$

$$\theta = -30^\circ + 180^\circ = 150^\circ$$

$$\boxed{6 \text{ cis } 150^\circ}$$

WHY

$$r = \sqrt{x^2 + y^2}.$$

$\tan^{-1}$ only gives QI or QIV.

$x < 0, y > 0$ is QII: add 180° .

Convert to rectangular form. (3 points)

11. $4 \operatorname{cis} \frac{5\pi}{4}$

WORK

$$4 \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) = 4 \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right)$$

$$= \boxed{-2\sqrt{2} - 2\sqrt{2}i}$$

Trap: A finished answer is not an equation: $-2\sqrt{2} - 2\sqrt{2}i$ cannot be divided down to $-1 - i$.

WHY

$\frac{5\pi}{4}$ is QIII: both negative.

Distribute the 4.

Using $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... is the following in the Julia set? Show ALL work. (3 points)

12. $z = -1$

YES / NO

WORK

$$(-1)^2 - 1 = 0$$

$$0^2 - 1 = -1, (-1)^2 - 1 = 0, \dots$$

never exceeds 2:

WHY

First step; $|0| = 0$.

Cycles $0, -1, 0, -1$.

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

13. $\frac{5\sqrt{2}(\cos 60^\circ + i \sin 60^\circ)}{10(\cos 285^\circ + i \sin 285^\circ)}$

WORK

$$\frac{5\sqrt{2}}{10} \operatorname{cis}(60^\circ - 285^\circ) = \frac{\sqrt{2}}{2} \operatorname{cis}(-225^\circ)$$

$$= \frac{\sqrt{2}}{2} \operatorname{cis} 135^\circ$$

$$= \frac{\sqrt{2}}{2} \left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right)$$

$$= \boxed{-\frac{1}{2} + \frac{1}{2}i}$$

Trap: Subtracting the angles can go negative. Add 360° before you look up cos and sin.

WHY

Quotient theorem: divide the r 's, subtract the θ 's.

Negative angle: add 360° .

135° is QII.

$$\sqrt{2} \cdot \sqrt{2} = 2.$$

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

14. $\left[\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \right] \left[4 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) \right]$

WORK

$$\sqrt{2} \cdot 4 \operatorname{cis} \left(\frac{3\pi}{4} + \frac{6\pi}{4} \right) = 4\sqrt{2} \operatorname{cis} \frac{9\pi}{4}$$

$$\frac{9\pi}{4} - 2\pi = \frac{9\pi}{4} - \frac{8\pi}{4} = \frac{\pi}{4}$$

$$4\sqrt{2} \operatorname{cis} \frac{\pi}{4} = 4\sqrt{2} \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right)$$

$$= \boxed{4 + 4i}$$

Trap: Subtracting π instead of 2π lands in the opposite quadrant and flips both signs.

WHY

Product theorem; $\frac{3\pi}{2} = \frac{6\pi}{4}$.

Past 2π : subtract 2π , not π .

$\frac{\pi}{4}$ is QI.

$$\sqrt{2} \cdot \sqrt{2} = 2.$$

Using $z, z^2 + z, (z^2 + z)^2 + z, \dots$ is the following number in the Mandelbrot set? Show ALL work. (3 points)

15. $z = 1 + i$

YES / NO

WORK

$$z = 1 + i, |z| = \sqrt{2}$$

$$z^2 + z = 2i + 1 + i = 1 + 3i$$

$$|1 + 3i| = \sqrt{1 + 9} = \sqrt{10} \approx 3.16 > 2$$

NO

WHY

Start.

$$(1 + i)^2 = 1 + 2i + i^2 = 2i.$$

Exceeds 2.

Find each power. Write your answer in rectangular form. (3 pts)

16. $(-1 - i)^6$

WORK

$$r = \sqrt{1 + 1} = \sqrt{2}, \theta = \tan^{-1}(1) = 45^\circ \rightarrow 225^\circ$$

$$(\sqrt{2})^6 \text{cis}(6 \cdot 225^\circ) = 8 \text{cis} 1350^\circ$$

$$1350^\circ - 3(360^\circ) = 1350^\circ - 1080^\circ = 270^\circ$$

$$8(0 - 1i) = \boxed{0 - 8i}$$

WHY

$x < 0, y < 0$: QIII, add 180° .

$$(\sqrt{2})^6 = 2^3 = 8.$$

Subtract full turns until it fits.

$$\cos 270^\circ = 0, \sin 270^\circ = -1.$$

Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

17. $x^4 - (-8 + 8i\sqrt{3}) = 0$

WORK

$$x^4 = -8 + 8i\sqrt{3}$$

$$r = \sqrt{64 + 192} = 16, \theta = \tan^{-1}(-\sqrt{3}) = -60^\circ \rightarrow 120^\circ$$

$$\sqrt[4]{16} = 2, \alpha = \frac{120^\circ + 360^\circ k}{4} = 30^\circ + 90^\circ k$$

$$\boxed{2 \text{cis} 30^\circ, 2 \text{cis} 120^\circ, 2 \text{cis} 210^\circ, 2 \text{cis} 300^\circ}$$

Trap: Move the number to the other side first; its sign changes.

WHY

Isolate x^4 .

$x < 0, y > 0$: QII, add 180° .

$$k = 0, 1, 2, 3.$$

Each is $2(\cos \alpha + i \sin \alpha)$; trig form, stop here.

Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points)

18. $(-8, \frac{3\pi}{4})$

WORK

$$x = -8 \cos \frac{3\pi}{4} = -8 \left(-\frac{\sqrt{2}}{2}\right) = 4\sqrt{2}$$

$$y = -8 \sin \frac{3\pi}{4} = -8 \left(\frac{\sqrt{2}}{2}\right) = -4\sqrt{2}$$

$$\boxed{(4\sqrt{2}, -4\sqrt{2})}$$

Trap: Dropping the negative on r gives $(-4\sqrt{2}, 4\sqrt{2})$, the mirror point.

WHY

Keep r 's negative sign in both lines.

$\frac{3\pi}{4}$ is QII: cosine negative, sine positive.

Check: negative r puts the point opposite $\frac{3\pi}{4}$, in QIV. It is.

Convert the rectangular coordinates to polar coordinates with $r > 0$ and $0^\circ \leq \theta < 360^\circ$. (3 points)

19. $(3, -3\sqrt{3})$

WORK

$$r = \sqrt{3^2 + (-3\sqrt{3})^2} = \sqrt{9 + 27} = 6$$

$$\tan \theta = \frac{-3\sqrt{3}}{3} = -\sqrt{3}: \text{reference angle } 60^\circ$$

$$\theta = 360^\circ - 60^\circ = 300^\circ$$

$$(6, 300^\circ)$$

Trap: $\tan^{-1}(-\sqrt{3}) = -60^\circ$ is outside $[0^\circ, 360^\circ)$. Add 360° .

WHY

$$r = \sqrt{x^2 + y^2}.$$

$$\tan \theta = \frac{y}{x}.$$

$x > 0, y < 0$: QIV.

Convert the rectangular equation to a polar equation. Solve for r . (3 points)

20. $5x + 2y = 8$

WORK

$$5r \cos \theta + 2r \sin \theta = 8$$

$$r(5 \cos \theta + 2 \sin \theta) = 8$$

$$r = \frac{8}{5 \cos \theta + 2 \sin \theta}$$

WHY

$$x = r \cos \theta, y = r \sin \theta.$$

Factor out r .

Divide.

Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points)

21. $(-3, 0^\circ)$ and $(6, 90^\circ)$

WORK

$$(-3, 0^\circ) \rightarrow (-3, 0), (6, 90^\circ) \rightarrow (0, 6)$$

$$m = \frac{6 - 0}{0 - (-3)} = 2, y = 2x + 6$$

$$2x - y = -6$$

$$2r \cos \theta - r \sin \theta = -6$$

$$r = \frac{-6}{2 \cos \theta - \sin \theta}$$

WHY

Negative r at 0° is on the left.

Slope, y -intercept 6.

Standard form.

Substitute.

Factor r , divide. Same as $\frac{6}{\sin \theta - 2 \cos \theta}$.

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

22. $r = -8 \sin \theta$

WORK

$$r^2 = -8r \sin \theta$$

$$x^2 + y^2 = -8y$$

$$x^2 + y^2 + 8y + 16 = 16$$

$$x^2 + (y + 4)^2 = 16$$

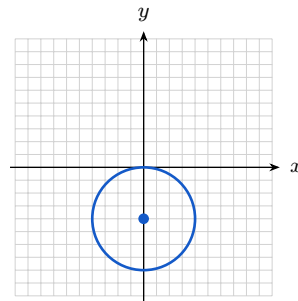
WHY

Multiply both sides by r .

$$r^2 = x^2 + y^2, r \sin \theta = y.$$

Complete the square: add 16 to both sides.

Circle, center $(0, -4)$, radius 4.



Trap: The minus sign moves the center below the pole, not above it.

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

23. $r = \frac{5}{\cos \theta + \sin \theta}$

WORK

$$r(\cos \theta + \sin \theta) = 5$$

$$r \cos \theta + r \sin \theta = 5, x + y = 5$$

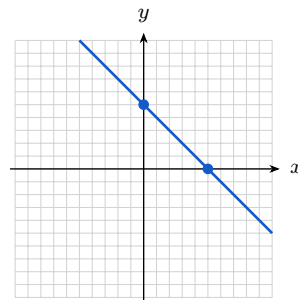
$$y = -x + 5$$

WHY

Clear the fraction.

Distribute, then substitute.

Line: slope -1 , both intercepts 5.



Find the polar coordinates of the points of intersection for the given interval of θ . (4 points)

24. $r = 3 + 2 \cos \theta$, $r = 3 - 2 \sin \theta$
 $0 \leq \theta < 2\pi$

WORK

$$3 + 2 \cos \theta = 3 - 2 \sin \theta, \sin \theta = -\cos \theta$$

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\frac{3\pi}{4}: r = 3 + 2 \left(-\frac{\sqrt{2}}{2} \right) = 3 - \sqrt{2}; \quad \frac{7\pi}{4}: r = 3 + \sqrt{2}$$

$$\left(3 - \sqrt{2}, \frac{3\pi}{4} \right), \left(3 + \sqrt{2}, \frac{7\pi}{4} \right)$$

WHY

Set $r = r$; the 3s cancel.

Unit-circle points where $y = -x$.

Plug each back in.

Neither curve reaches the pole ($r \geq 1$).

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

25. $r^2 = 16 \cos 2\theta$

WORK

lemniscate

WHY

r^2 equation: figure 8, loops 4 long along the x -axis.

Trap: $r^2 = a^2 \cos 2\theta$ or $r^2 = a^2 \sin 2\theta$ is the lemniscate. ($r^2 = 16$ alone would be a circle.)